



Karan Bansal, Founder (right), and Tanya Goyal, Co-founder (left), displaying their product

in the inlet space, which is divided into two parts: one part for air intake and the other for the biodegradable filter,” explains Bansal.

The product operates at a lower clean air delivery rate (CADR) and efficiency compared to a regular purifier. “The idea is to maintain the pollution level below a certain threshold over time, rather than rapidly removing it within a short duration, by merging the airflow from the purifier with that of the fan to save energy. This way we don’t need to use a HEPA filter, which is used in a traditional air purifier, to push the entire air volume through it, causing a significant pressure drop. Instead, we use a biodegradable filter to keep both functions running concurrently without using excessive energy for air purification,” elaborates Bansal.

By using CFD software simulations, Karban’s R&D team, which is well-versed in technology along with CAD and industrial design, reduced the design optimisation time compared to traditional methods, where physical prototypes are built and tested iteratively. “CFD software allows us to create a CAD model of the geometry or housing and simulate airflow interactions. By making iterative changes to the design, we can optimise airflow without relying solely on physical prototypes,” he reveals.

Despite all these functionalities, Karban Airzone claims to be an energy-efficient appliance. Each side of the Karban Airzone (each blower) consumes around 28 watts of elec-

tricity at the maximum setting. The light operates independently and is a standard LED, providing around 25 watts of power and emitting around 2000 lumens.

The startup sources components from various Indian vendors, with assembly conducted in Karban’s own manufacturing facility, and is currently looking to expand its distributor network. The startup is also seeking potential collaborations with production and sales vendors/partners. The team consists of more than 15 full-time and part-time employees, mainly in engineering, manufacturing, and sales.

With seed funding from CII’s Startup India, the startup has recently raised \$1.07 million in a seed round led by Titan Capital, Rainmatter, and All In Capital. Among the other investors in the seed round were JK Family Office and Urban Company’s Anupam Mittal. A few other well-known investors are Sarang Iyengar, Karan Jindal, Atul Gupta, and Abhishek Goyal.

Initially produced with wood due to cost and scalability considerations, Karban plans to develop a plastic version. With all its operations in Jaipur due to lower expenses and favourable sourcing opportunities, Karban remains open to relocating or expanding manufacturing facilities as needed in the future.

—Yashasvini Razdan

3 chargeMOD OFFERS ONE-STOP SOLUTION FOR WHOLE EV ECOSYSTEM

Kerala-based energy tech startup chargeMOD Pvt Ltd began its operations in 2021-22 and initiated monetisation in early 2022.

Founded in 2019, chargeMOD specialises in developing energy technologies, including electric vehicle (EV) charging, power electronics, control and communication architecture, postmark energy tech

hardware, and grid-side technologies.

Ramanunni M, Advait C, Anoop V, Visakh V Raj, and Mithun Krishnan M are the five founders of this startup. Due to the pandemic, the startup began operational work

in 2021-22 and initiated monetisation in early 2022.

Initially, chargeMOD aimed to connect EV drivers with charging stations, similar to Uber’s model. However, realising the high hardware costs of EV drivers, which were not manufactured in India, the company refocused on designing core technology and IP. This